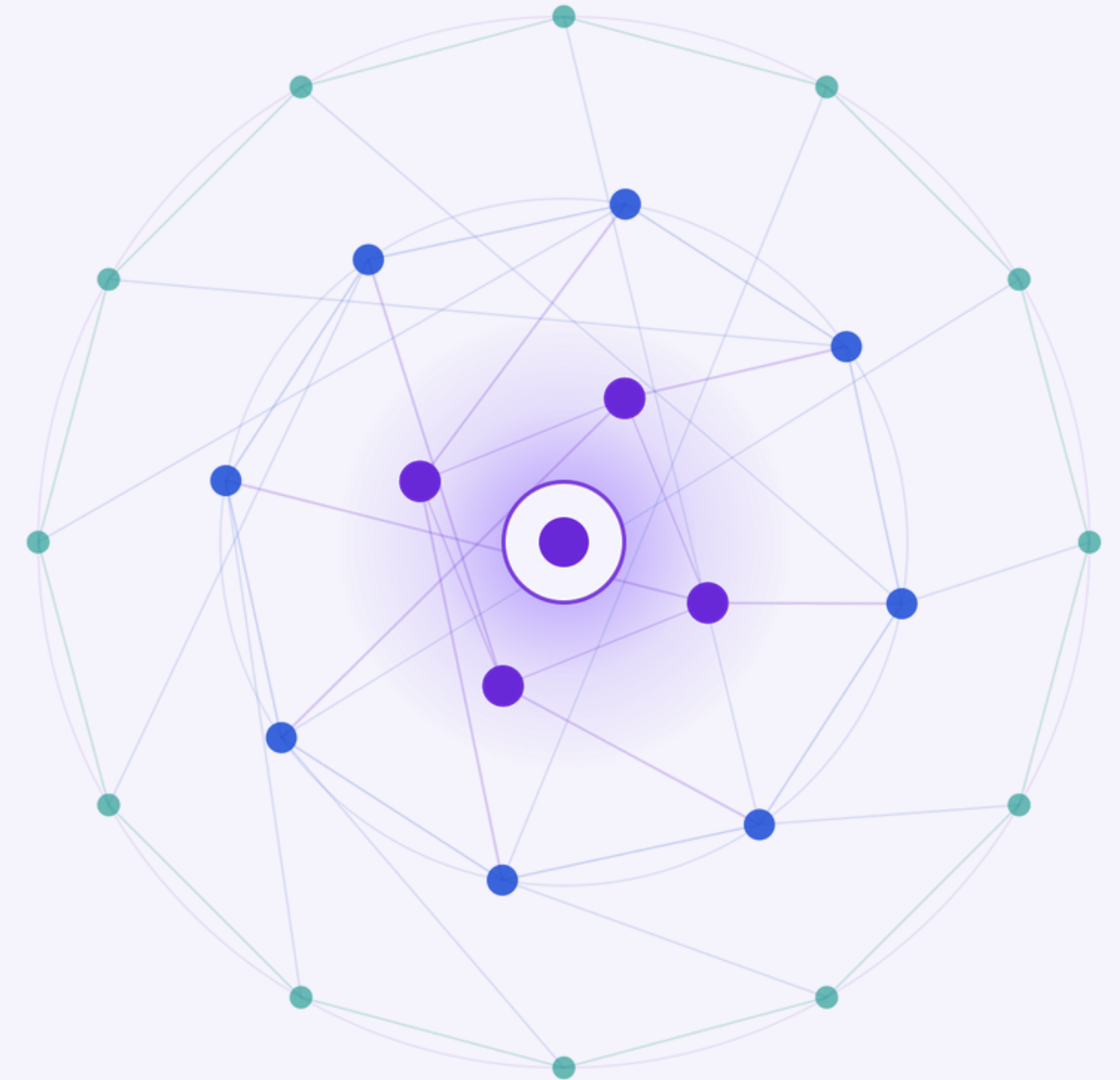


SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - STREAMING LIVE RAG
- Team Name - SYNQ
- College Name - SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
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- Submission Github link – <https://github.com/Banisher2005/AVY>



02. Theme

Theme 04 – Streaming Live RAG

Make Retrieval-Augmented Generation work in **real-time, natural conversations**.



Theme Overview

Build a system that can process a **live, continuously streaming** conversation, intelligently decide when to retrieve information, handle **multiple intents** within an utterance, and provide **grounded, citation-backed answers** with low latency.



The Problem

Traditional RAG systems assume a **clean, complete search query**. But in real conversations, users speak naturally, with **incomplete information, multiple questions, follow-ups** and **refinements**. Retrieving too early can give irrelevant results, while waiting too long increases latency and breaks the **conversational flow**.



Key Challenges in Streaming RAG



Incremental Input

Information arrives in chunks, not a full query.



Multiple Intents

One utterance can contain several distinct questions.



Retrieval Decision

Should we WAIT, RETRIEVE, or NOT RETRIEVE?



Follow-up Refinement

New details should refine the existing answer, not restart from scratch.



Example: A Natural Spoken Request



User (Chunk 1)

"Tell me about Samsung's..."



AVY Decision



WAIT

Query is incomplete



User (Chunk 2)

"...latest processor."



AVY Decision



RETRIEVE

Enough information for meaningful search



User (Chunk 3)

"...and compare it with the previous generation."



AVY Decision



RETRIEVE & DECOMPOSE

Multiple intents detected



User (Follow-up)

"Now compare its battery efficiency as well."



AVY Decision



REFINE

Use previous context and update the answer



Our Solution – AVY

AVY is a **streaming, conversation-aware RAG assistant** that can detect retrieval needs, decompose complex utterances, retrieve and fuse relevant information, and refine answers in **real-time**, providing **grounded, citation-backed responses** with **low time-to-first-token**.



03. Existing Solutions & Gaps

Why Traditional RAG Struggles with Live Conversation

Traditional RAG is designed for completed queries. But real users speak naturally — with incomplete information, multiple intents and follow-ups. This creates key challenges.



01 Incomplete Queries

Users don't always give a complete query. Information arrives gradually as they speak.



Traditional RAG

Waits for a complete and well-formed query before retrieval.

"Tell me about Samsung's..."

No retrieval yet



Live Conversation (User Behaviour)

The user speaks in chunks, with incomplete information.

"Tell me about Samsung's..."

...latest processor..."

...and compare it..."



AVY Solution

WAIT → RETRIEVE

Intelligently decides when to wait for more information and when to start retrieval based on semantic completeness.

02 Multiple Intents

A single utterance can contain multiple questions or information needs.



Traditional RAG

Treats the entire utterance as one search query.

"Compare Samsung's latest processor with the previous generation and explain the battery efficiency."



Live Conversation (User Behaviour)

One sentence may include several distinct intents.

- 1 Latest processor details
- 2 Previous generation details
- 3 Battery efficiency comparison



AVY Solution

Decompose → Retrieve → Combine

Breaks the utterance into multiple search-ready sub-queries, retrieves relevant information for each, and fuses the results into a single grounded answer.

03 Follow-up & Refinement

Users often add new details, corrections or follow-up questions that refine the original request.



Traditional RAG

Treats follow-ups as new, independent queries.

"Tell me about the processor."

"Now compare its battery efficiency."



Live Conversation (User Behaviour)

New information often builds on the previous context.

"Now compare its battery efficiency as well."



- Refers to previous intent
- Adds new information
- Updates the existing request



AVY Solution

Remember → Refine → Answer

Uses session context, previous intent, retrieved evidence and follow-up details to refine and update the answer instead of starting from scratch.



Key Takeaway

The gap isn't just better retrieval — it's making retrieval respond to how conversation actually unfolds.

From AVI to AVY – Streaming Live RAG

AVY is built on the AVI architecture and extends it with a complete Retrieval-Augmented Generation (RAG) layer to handle **real-time, multi-intent, conversational requests**.

1. AVI – Existing Architectural Foundation

AVI provides the core conversational architecture for routing, context management, provider abstraction and streaming.



Router

Request routing & classification



Context Subsystem

Conversation history & session handling



Provider Abstraction

LLM interface & model configuration



Streaming / Gateway

Token streaming, request lifecycle & error handling

Adapt
and
Extend

2. AVY – Streaming Live RAG

AVY adapts the AVI components and adds a complete RAG layer to enable **real-time, multi-intent, grounded and streaming responses**.



Streaming Input

Transcript chunks from user speech



Retrieval Controller

Decides
WAIT /
RETRIEVE /
NO_RETRIEVE



Multi-Intent Decomposition

Breaks the
utterance into
multiple
sub-queries



Retrieval & Fusion

Vector search,
deduplication
and reranking



Grounded Answer

Generates
response with
citations



Streaming Response

Token-by-token
answer with
low latency



Session-Aware Refinement

Uses previous context to
update answers



Citations & Traceability

Each claim is linked
to source documents

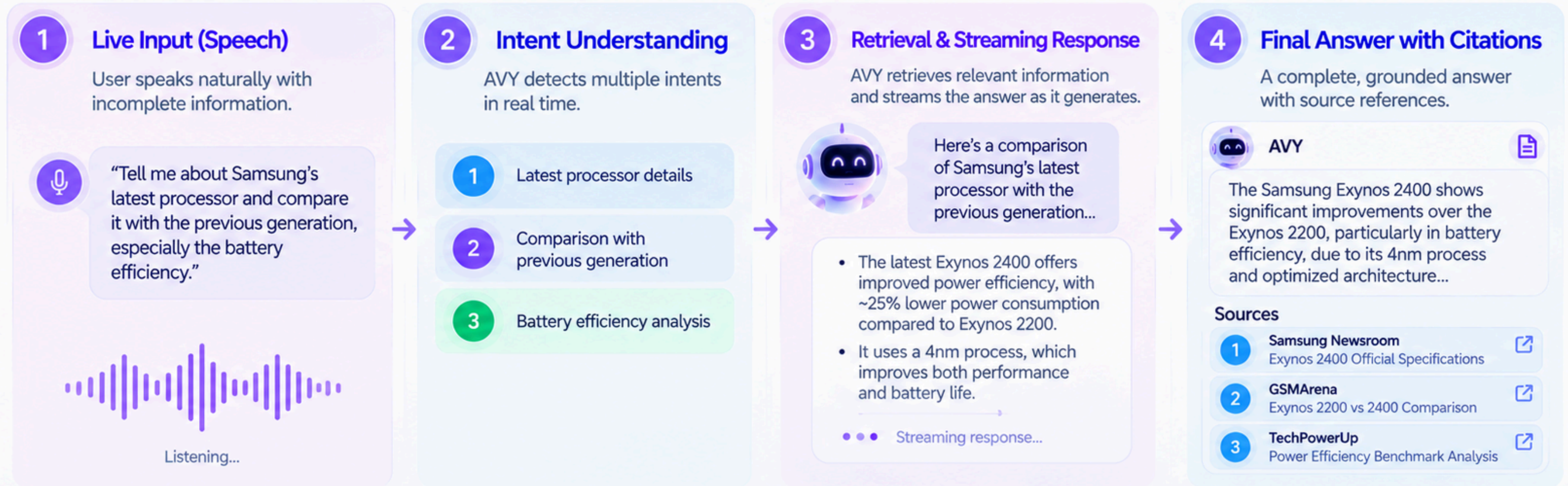


Telemetry & Evaluation

Measures recall,
groundedness, latency and
other key metrics

AVY in Action

A real-time example showing how AVY handles an evolving, multi-intent request — from live speech input to a grounded, streaming response with citations.



Key Takeaway from the Demo



Handles evolving, multi-intent queries



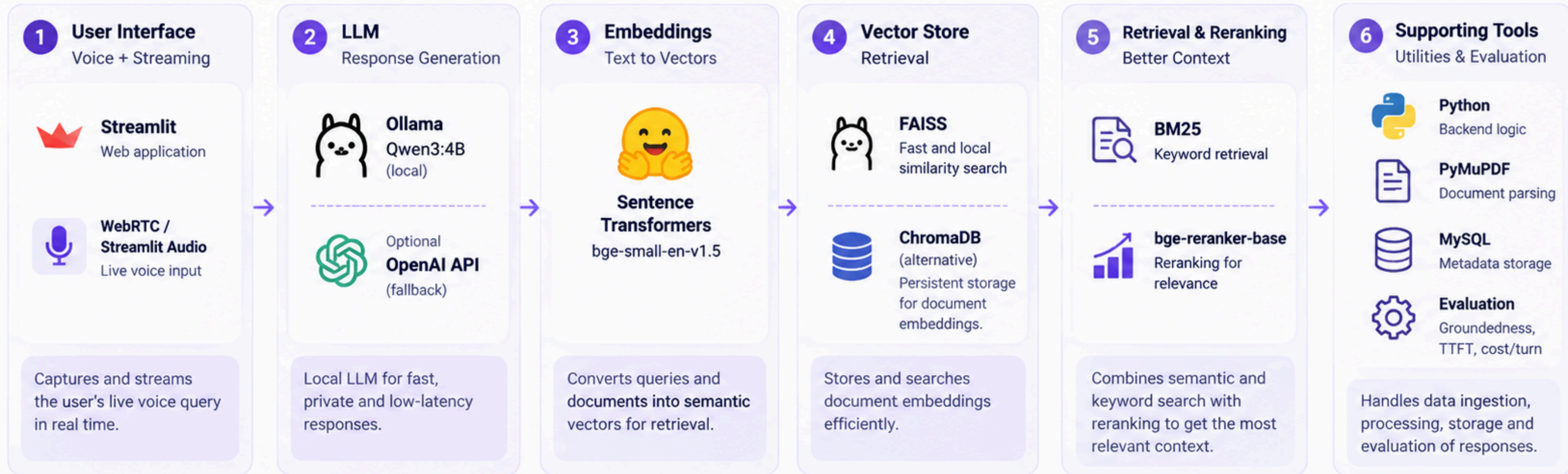
Provides real-time, streaming responses



Grounded with trusted sources and citations

Tools and Tech Stack Used

A lightweight, modular and privacy-focused stack to enable real-time, streaming RAG.



Local-First

Runs locally for privacy and low latency.



Modular

Components can be swapped or upgraded.



Real-Time

Optimized for streaming, interactive responses.

Innovation Highlights, Results & Limitations



INNOVATION HIGHLIGHTS

Key ideas that make AVY different

01



Streaming Voice Pipeline

Processes live speech and responds in real-time with low latency.

02



Adaptive Retrieval Controller

Decides when to wait, retrieve or not retrieve based on query completeness.

03



Multi-Intent Decomposition

Breaks complex queries into multiple search-ready intents.

04



Evidence Fusion & Reranking

Combines dense and sparse retrieval to improve relevance.

05



Grounded, Cited Responses

Generates trustworthy answers with source citations.



RESULTS WE WILL MEASURE

Key metrics to evaluate AVY



Retrieval Recall

How much relevant evidence is retrieved.



Answer Groundedness

How well the final answer is supported by sources.



TTFT (Time to First Token)

Time until the first streamed response is generated.



Retrieval Latency

Time spent retrieving and reranking evidence.



User Satisfaction

Relevance, accuracy and helpfulness of responses.



LIMITATIONS

Current scope and constraints



Transcript input may be simulated; real ASR is future scope.



Performance depends on corpus quality and available trusted sources.



Local LLM/reranker latency varies by machine.



Session-scoped context; no persistent cross-session user profile.

What's Next

From prototype to a robust streaming assistant

01



Real ASR Integration

- Replace simulated transcript input with real-time speech-to-text.
- Support continuous streaming input with low latency.

02



Adaptive Retrieval Controller

- Improve WAIT / RETRIEVE / NO_RETRIEVE using semantic completeness, session context and latency.
- Make retrieval decisions more context-aware and robust.

03



Hybrid Retrieval

- Combine dense + sparse retrieval to improve evidence coverage.
- Use vector search and keyword search for better recall and relevance.

04



Ablation & Evaluation

- Measure the contribution of decomposition, fusion, reranking and refinement independently.
- Evaluate on relevance, groundedness, latency and user satisfaction.

05



Deployment Hardening

- Pin dependencies, containerize the stack and provide a reproducible one-command demo.
- Prepare for real-world usage with monitoring and error handling.



LONG-TERM VISION

A seamless, real-time and reliable voice assistant for knowledge access.



Microphone
User speaks naturally



Streaming ASR
Real-time transcription



AVY Retrieval Controller
Decides, retrieves and refines



Grounded RAG
Uses trusted sources



Streaming Answer
Relevant, cited and real-time

Beyond the Basics

Additional features and impact areas that make AVY more practical, responsible and impactful.

01 Real-Time & Low Latency

Designed for live, natural conversations with fast, streaming responses.

- ✓ Token-by-token generation (TTFT)
- ✓ Optimized retrieval pipeline
- ✓ Handles evolving inputs smoothly



02 Multi-Intent Handling

Understands and processes multiple information needs in a single utterance.

- ✓ Intent decomposition
- ✓ Parallel sub-query retrieval
- ✓ Fused, coherent answer



03 Grounded & Trustworthy Answers

Every claim is linked to verifiable sources.

- ✓ Source citations with links
- ✓ Evidence highlighting
- ✓ Reduces hallucinations



04 Session-Aware Refinement

Remembers context and adapts as the conversation progresses.

- ✓ Tracks conversation history
- ✓ Incorporates follow-ups
- ✓ Refines and updates answers



05 Responsible & Safe

Built with guardrails for secure and reliable usage.

- ✓ Filters irrelevant or harmful content
- ✓ Respects privacy and sensitive data
- ✓ Provides transparent sources



06 Scalable & Extensible

Modular design to support new data sources and domains.

- ✓ Plug-and-play knowledge sources
- ✓ Works across domains (devices, specs, support, news)
- ✓ Can integrate with multimodal inputs (text, images, docs)



Key Takeaway

These enhancements make AVY not just a demo, but a **practical, trustworthy and future-ready solution.**



Thank You

— QUESTIONS? —